



QUALITY CIRCLE- (8D Approach)

PROJECT NAME

: Reducing power consumption by eliminating Vacuum Pump & 60 Hp motor

PLANT(Location)

: Podanur Works

DATE OF START OF PROJECT

: March 2018

DATE OF COMPLETION OF PROJECT

: April 2018

Project Title & Team Formation



Project Title: Reducing power consumption by eliminating vacuum pump & 60hp motor.

Implemented Area: Sheeting Machine -1 Shop floor, Podanur Works, Coimbatore

Cross Functional Team:



Mr. M. Palavesa Moorthy
Prod. Shift Incharge



Mr. M. Madan Raj
Prod. Shift Incharge



Team Leader
Mr. Y. Kesavaraj
SM1 Line Incharge



Mr. K. Senthil Kumar
Electrical Supervisor



Mr. N. Anil Kumar
QS & BE



Mr. Sivadasan
Machine Driver



Mr. Madhialagan
Machine Driver



Mr. Peter
Machine Driver

**Reducing Power consumption by eliminating 1 vacuum pump and 60Hp motor.
(Target:- To use 2 vacuum Pumps)**

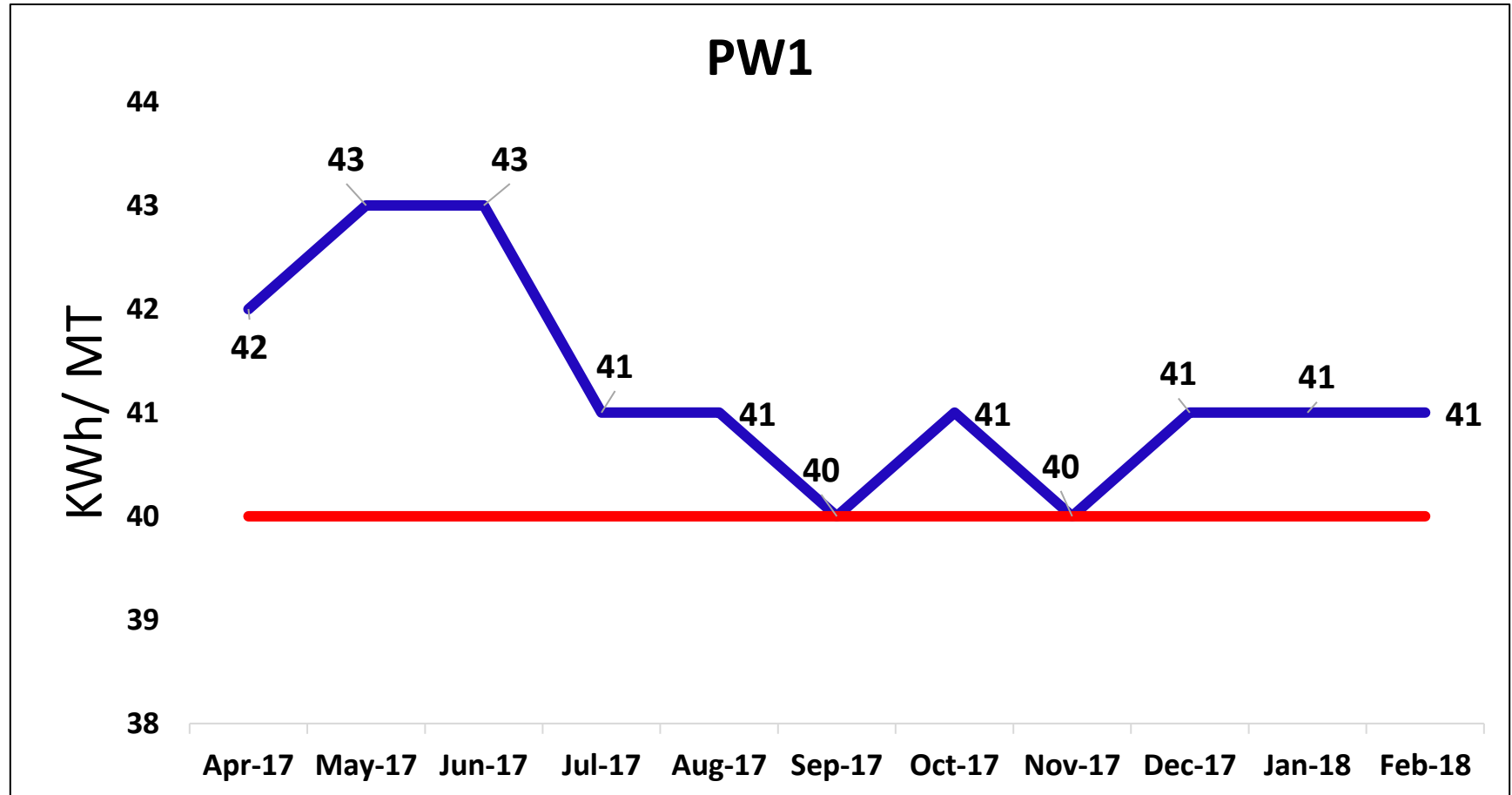
Reason for selecting THEME :

- *High power consumption.*
- *More drying of the slurry film.*
- *To reduce consumption of Bearings & rollers in Top Vacuum trays.*

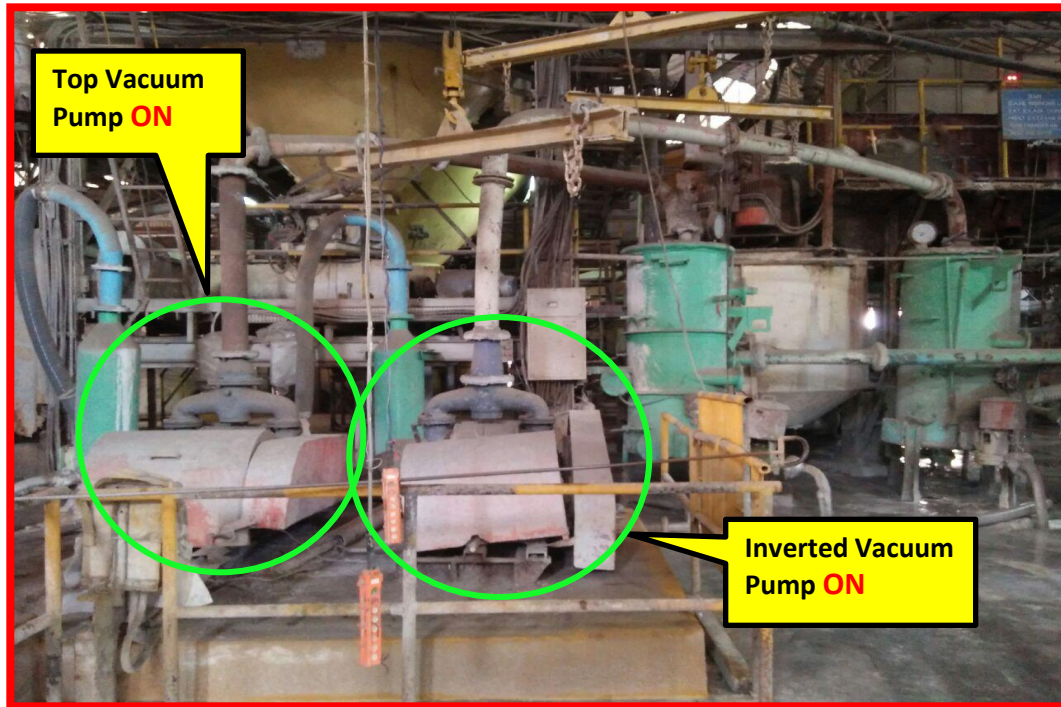
Problem :

More power consumption. If we reduce Vacuum below the optimum requirement we are facing problems like Layer falling from felt, Layer transferring problem to felt & Vat over concentration.

Existing Data – PW1 Power Consumption



Problem Description: Before situation



- Bottom suction Trays openings are more. So to maintain required vacuum intensity 1 dedicated vacuum pump is used for bottom suction trays.
- Due to use of Rollers with bearing arrangement for decreasing the load on the main drive there is less opening area for Top suction Trays. To compensate the vacuum requirement for drying the film we are using 3 Top suction trays and using 1 vacuum pump

Problem Description: Before Situation Process



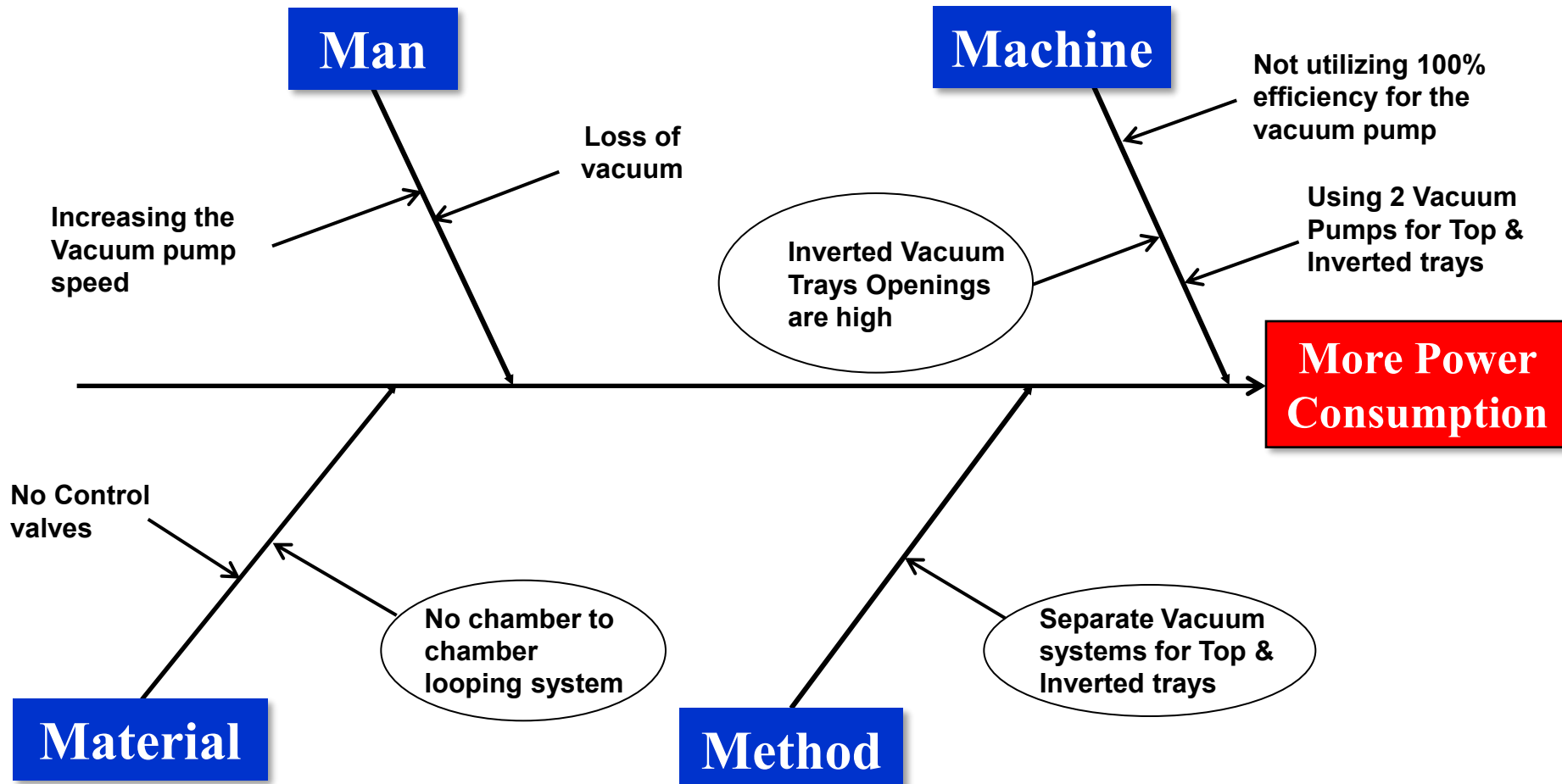
- In this, you can see the flow of Vacuum through the pipe lines & vacuum chambers towards Top & Inverted vacuum pumps when both the vacuum pumps are in **ON** Condition.

STEP – 3 INTERIMN CONTAINMENT ACTIONS



S.NO	CONTAINMENT ACTION	RESPONSIBILITY	TARGET DATE
1.	Reducing the Vacuum intensity	Prod. Shift In-charges / Line In-charge	Completed
2.	Collecting data & ideas for reducing the power consumption	Shift In-charges / Line In-charge / Electrical Dept	Completed

Cause & Effect Analysis

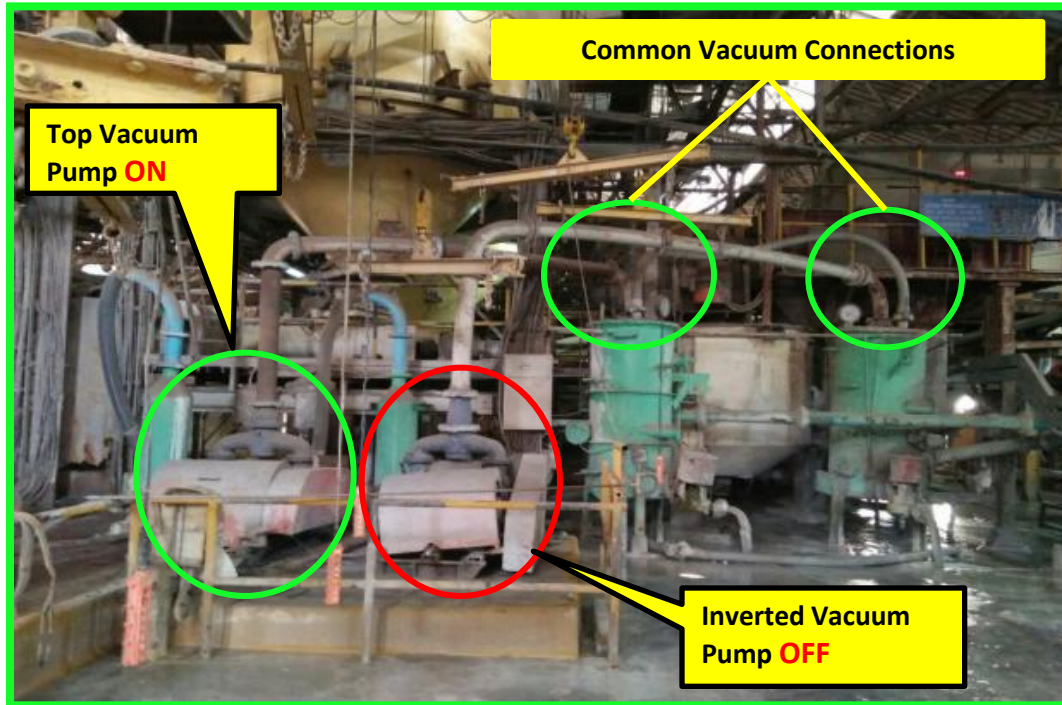


Problem Statement	More power consumption by vacuum pumps
1st why?	Why Power consumption by vacuum pumps is more?
Answer:	Using more vacuum
2nd why?	Why more vacuum is being used?
Answer:	Two Separate vacuum pumps are used for Inverted & Top trays.
3rd why	Why Two Separate vacuum pumps are used for Inverted & Top trays?
Answer:	Opening area of the inverted vacuum trays are high. To maintain required vacuum intensity separate vacuum pumps are used for inverted vacuum trays. And for Top trays 1 vacuum pump is used.
Root Cause:	Design constraint of the Inverted Vacuum Trays.
Solution	1. Reduce the opening area of Inverted vacuum trays.
	2. Provide looping arrangement between Top & Inverted vacuum chambers.

Permanent Corrective Actions

S. No	Actions	Resp.	Target Date	Status
1	Modify the opening area of Bottom vacuum trays by decreasing the opening area & volume of trays.	Line In-charge	30/03/18	Completed
2	Increase the opening area of the Top vacuum trays by removing the rollers & bearings in the trays and install Poly-picks.	Line In-charge	30/03/18	Completed
3	Remove one Top vacuum trays and to use only Two top trays.	Line In-charge	30/03/18	Completed
4	Provide looping arrangement b/w Top & Inverted vacuum chambers	Line In-charge	02/04/18	Completed

Permanent Corrective Actions



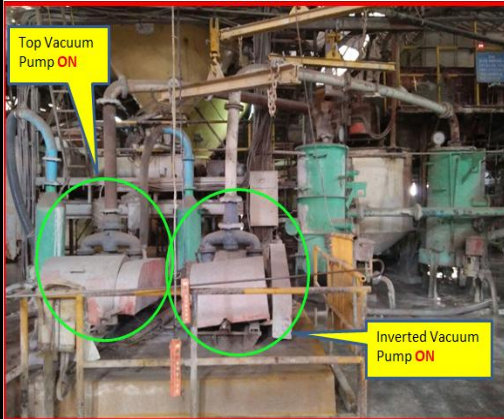
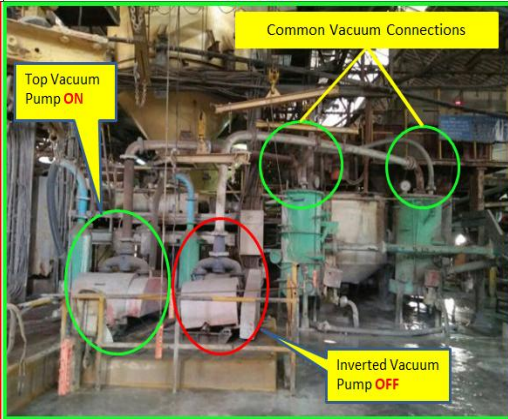
- By modifying the vacuum circuit we are now using only 1 vacuum pump for Top and Bottom trays instead of using Two vacuum pumps.
- By this we have stopped using 1 Vacuum pump, 60 Hp motor, Rollers & Bearings in vacuum trays & other consumables for maintenance.

After Situation Explanation



- In the above picture, Top vacuum pump is **ON** bottom vacuum pump is **OFF**.
- Vacuum from Top vacuum trays flows towards Top Vacuum Pump through vacuum chamber & pipe lines.
- Vacuum from Inverted vacuum trays flows towards Top vacuum pump through pipe lines, then to Inverted vacuum chamber and then to Top vacuum chamber, avoiding the flow towards Inverted vacuum pump.

Permanent Corrective Actions

KAIZEN SHEET	Department	Production	Works	BW	SW	CW	KW	LW	PW	NW	BW- ESBS	
Kaizen Number	Line / Section	SM1	Result	P		Q		C	D	S	M	
PW/KZN/04/2018	Machine /Equipment	Vacuum Pump		Productivity		Quality		Cost	Delivery	Safety	Morale	
KAIZEN DESCRIPTION : Usage of only one vacuum pump for Top & Inverted vacuum trays instead of two vacuum pumps.			IDEA / Thought : To use only one vacuum pump for Top & Inverted vacuum trays.									
Problem / Present Status		Counter Measure		Target								
Previously for Top & Inverted vacuum separate vacuum pumps are used. (60 Hp + 60 Hp) = 120 Hp.		By connecting common vacuum line from chamber to chamber with looping arrangement we switched off 1 vacuum pump of 60Hp and using only one vacuum pump for Top & Inverted vacuum trays.		Kaizen Start Date			03 April 2018					
				Kaizen Finish Date			03 April 2018					
BEFORE		AFTER		Team Leader								
				Y. Kesavaraj (SM1 Line Incharge)								
				Team Members :								
				M. Palavesa Moothy (Shift officer), M. Madan Raj (Shift Officer), Sivadasan(Machine Driver), Peter(Machine Driver), Madhialagan(Machine Driver), K. Senthil Kumar(Electrical Supervisor)								
				Investment cost :			Rs. 5000/-					
				Benefits :			Motor			Drive		
				Power saving = $\sqrt{3} \times V \times I \times \text{Cos}\Phi$						Power saving = $\sqrt{3} \times V \times I \times \text{Cos}\Phi$		
				= $(\sqrt{3} \times 415 \times 60 \times 0.8) / 1000$						= $(\sqrt{3} \times 415 \times 5 \times 0.8) / 1000$		
				= 34.502 x 24 hrs x 28 Production days						= 2.875 x 24 hrs x 28 Prod. Days		
				= 23,185 Units/Month						= 1932 Units/Month		
				Power saving per Month = (23185 + 1932) Units x 8 Rs. Per Unit = Rs. 2,00,936 /-								
				Power saving per Year = Rs. 24,11,232 /- (Around 24 Lakhs)								
				KAIZEN SUSTAINANCE :								
				WHAT TO DO			Regular Inspection and Maintenance.					
				HOW TO DO			Training and Instructions need to given to the Process persons.					
				FREQUENCY			Weekly					
				Scope Of Horizontal Deployment								
				BW-ESBS	NW	BW	CW	KW	LW	PW	SW	
				✓	✓	✓	✓	✓	✓	✓	✓	
				Equipment		Target Date				Resp	Status	
Why Why Analysis		GRAPHS										
Problem :- Power consumption of vacuum pumps is more.		Power consumption in Units per 1000 M²N										
Why 1. From previously we are using separate vacuum pumps of 60Hp each for both the top & inverted vacuum trays.												
Why 2. There is no common vacuum line from chamber to chamber with looping arrangement.												
Why 3.												
Why 4.												
Why 5.												
Action Taken : By connecting common vacuum line from vacuum chamber to chamber we switched off 1 vacuum pump of 60Hp.												

Why Why Analysis

Problem :- Power consumption of vacuum pumps is more.

Why 1. From previously we are using separate vacuum pumps of 60Hp each for both the top & inverted vacuum trays.

Why 2. There is no common vacuum line from chamber to chamber with looping arrangement.

Why 3.

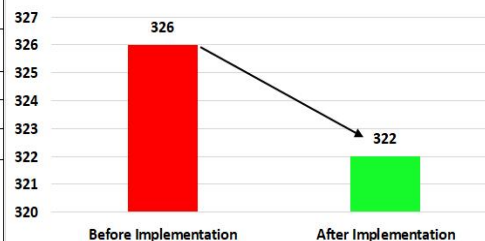
Why 4.

Why 5.

Action Taken : By connecting common vacuum line from vacuum chamber to chamber we switched off 1 vacuum pump of 60Hp.

GRAPHS

Power consumption in Units per 1000 M²N



Calculation of Power consumption & Cost Savings of Vacuum Pumps:

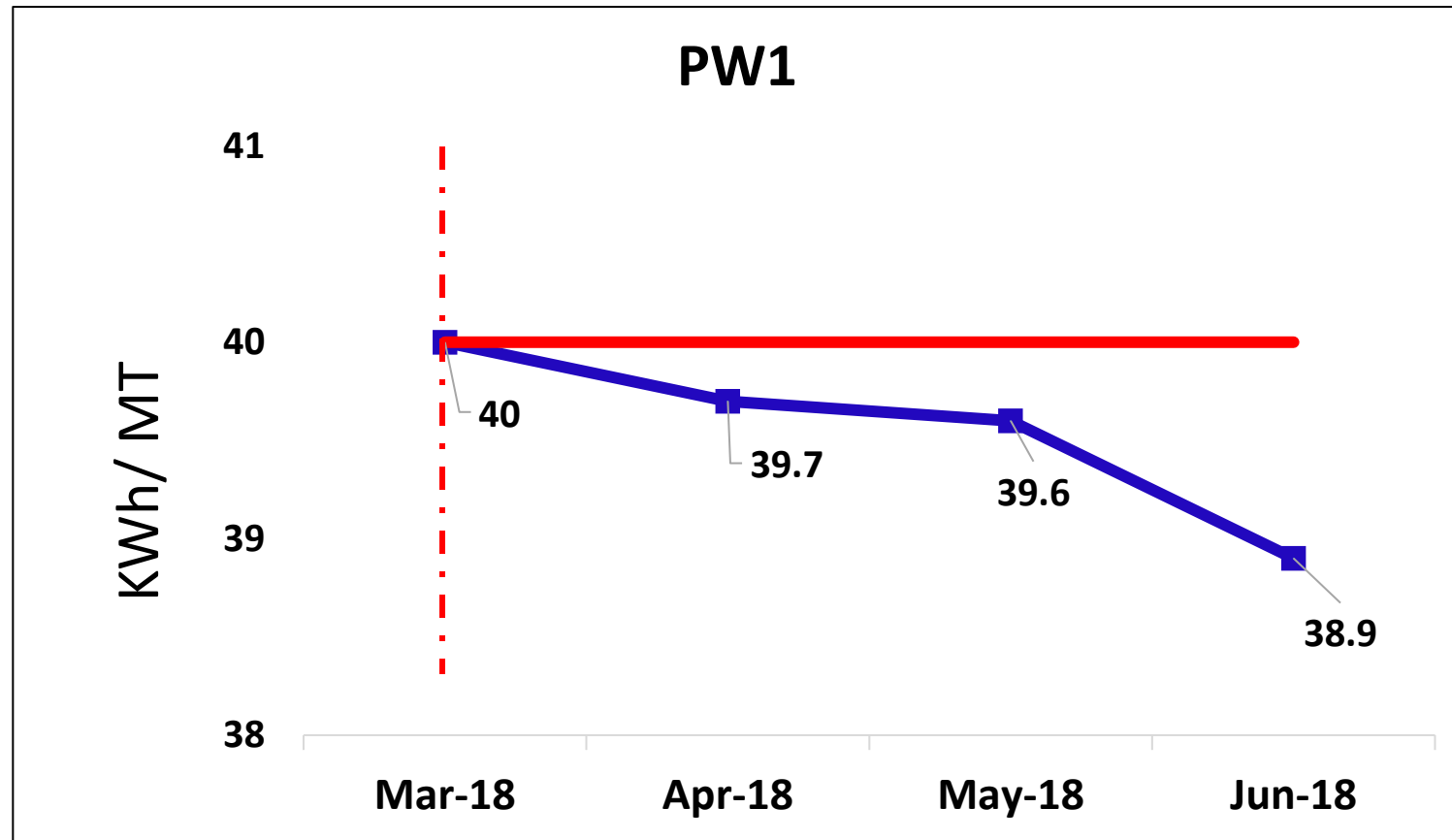
- Previously we are using 2 Vacuum Pumps of 60Hp & 1 Vacuum Pump of 40Hp.
- Now we are using 1 Vacuum Pump of 60Hp & 1 Vacuum Pump of 40Hp.

$$\begin{aligned}\text{Power Saving} &= \sqrt{3} \times V \times I \times \cos\phi \\ &= (\sqrt{3} \times 415 \times 70 \times 0.8) / 1000 \\ &= 34.502 \times 24 \text{ Hrs} \times 28 \text{ Production days} \\ &= \mathbf{27,049 \text{ Units/Month By eliminating 60Hp Vacuum Pump motor}}\end{aligned}$$

$$\begin{aligned}\text{Cost Saving per Month} &= \mathbf{27,049 \times 8 \text{ Rs. Per Unit}} \\ &= \mathbf{\text{Rs. 2,16,392/-}}\end{aligned}$$

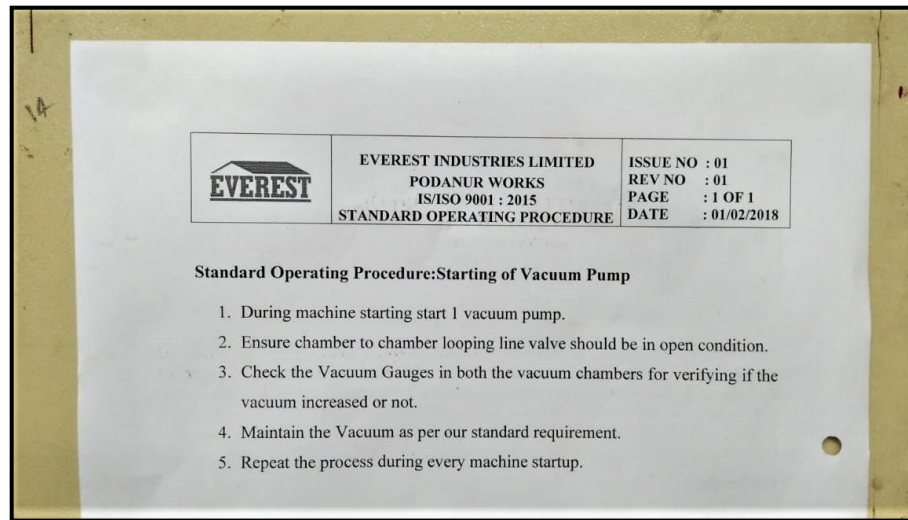
$$\text{Cost Saving per Year} = \mathbf{25.96 \text{ Lakh/Annum}}$$

- Cost Saving by eliminating Roller suction trays & Bearings = **1.72 Lakh/Annum**
- Cost Saving by eliminating Maintenance of 1 Vacuum Pump = **Rs. 17,000/Annum**
- **Total Cost Saving is 27.95 Lakh/Annum**



Tangible Benefits	Intangible Benefits
• Average Monthly power consumption of vacuum pumps is reduced from 71486 Units to 44437 Units.	• Morale increased due to reduction in power consumption.
• 37.83% reduction in Vacuum pumps power consumption.	• Morale increased due to reduction in effort for maintenance of vacuum pumps, rollers & bearings of Top vacuum trays.
• Use of rollers and bearings in Top vacuum trays is eliminated. And maintenance of them is eliminated.	
• Eliminated the use of 1 vacuum pump and motor of 60Hp.	
• Potential cost saving of 27.95 Lakh./Annum	

- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.



Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Congratulate the team

Yes / No	Signature / Title / Date	Findings
Yes	M. Gnana Prakash	Problem solving is an effective tool which is used for knowing the hidden facts of more power consumption. Further this process can be used in other area for solving the problems.

Thank You